

Name: _____

Show all work for credit. Circle your final answer.

1. Find the **sum**.

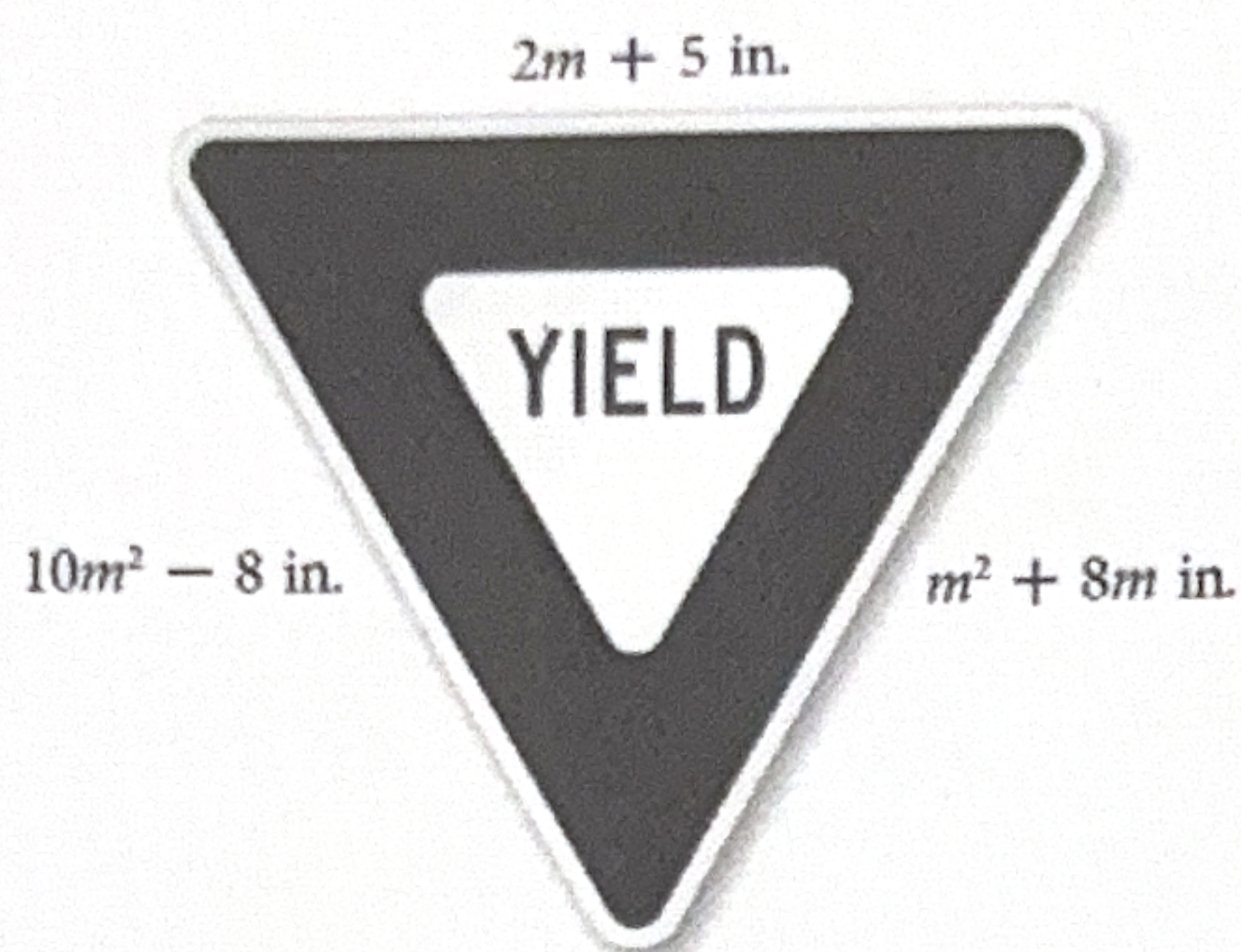
$$(5ab + 5ac - 9bc) + (-ab + 10ac + 4bc)$$

2. Find the **difference**.

$$(-12r^2s + 2rs^2 + 6rs) - (3r^2s + 8rs^2 - rs)$$

3. Find the **perimeter** of the triangular traffic sign

$$\text{Perimeter} = \text{Side} + \text{Side} + \text{Side}$$



4. The **revenue** for a company selling x products is modeled by the polynomial expression $(15x^3 + 6x^2 + 700)$ dollars. The operating **cost** for the company is modeled by the polynomial expression $(60x^2 + 2000)$ dollars. What simplified expression models the **profit** in dollars of the company as a function of x ?

$$\text{Profit} = \text{Revenue} - \text{Cost}$$

If the company sold 10 products, how much profit would they make? _____